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Original Article

Identification of CXCR4 inhibitory activity in natural compounds using cheminformatics-guided machine learning algorithms

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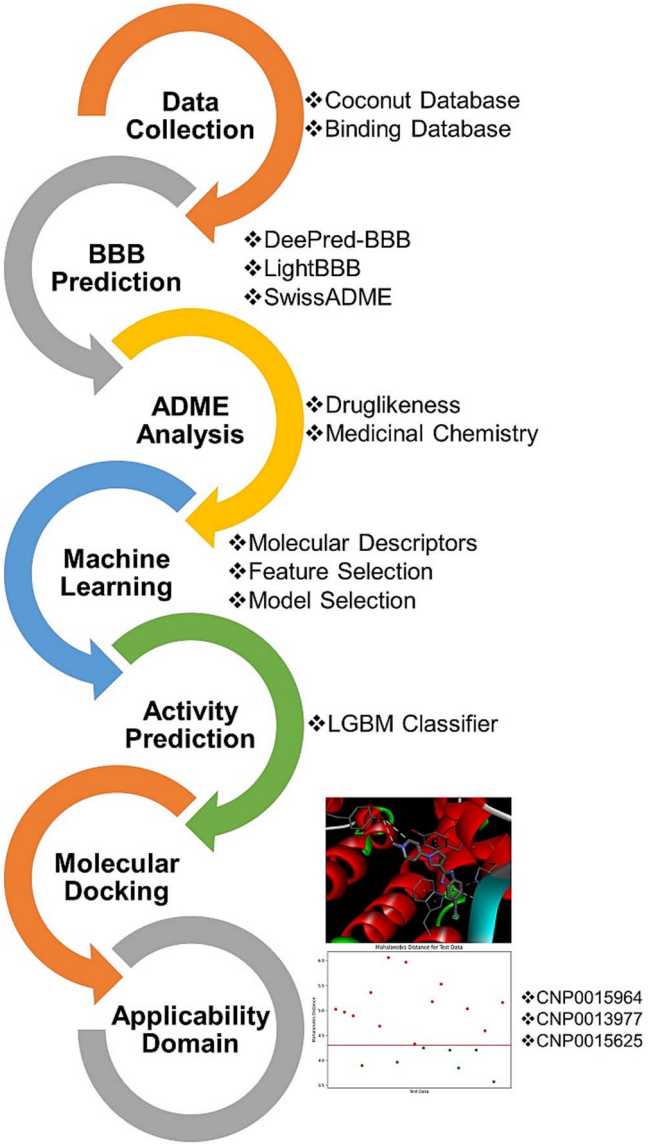
Abstract

Neurodegenerative disorders are characterised by progressive damage to neurons that leads to cognitive impairment and motor dysfunction. Current treatment options focus only on symptom management and palliative care, without addressing their root cause. In our previous study, we reported the upregulation of the CXC motif chemokine receptor 4 (CXCR4), in Alzheimer's disease (AD) and Parkinson's disease (PD). We reached this conclusion by analysing gene expression patterns of AD and PD patients, compared to healthy individuals of similar age. We used RNA sequencing data from Gene Expression Omnibus to carry out this analysis. Herein, we aim to identify natural compounds that have potential inhibitory activity against CXCR4 through cheminformatics-guided machine learning, to aid drug discovery for neurodegenerative disorders, especially AD and PD. Natural compounds are gaining prominence in the treatment of neurodegenerative disorders due to their biocompatibility and potential neuroprotective properties, including their ability to modulate CXCR4 expression. Recent advances in artificial intelligence (AI) and machine learning (ML) algorithms have opened new avenues for drug discovery research across various therapeutic areas, including neurodegenerative disorders. We aim to produce an ML model using cheminformatics-guided machine learning algorithms using data of compounds with known CXCR4 activity, retrieved from the Binding Database, to analyse various physicochemical attributes of natural compounds obtained from the COCONUT Database and predict their inhibitory activity against CXCR4.

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Graphical Abstract



Cheminformatics-guided machine learning methodology to analyse the physicochemical attributes of natural compounds and predict their inhibitory activity against CXCR4.

Insight Box
This work extends our previous study published in Integrative Biology (DOI: 10.1093/intbio/zyad012). We aim to demonstrate the effectiveness of AI and ML in identifying potential treatment options for Alzheimer's and Parkinson's diseases. By analysing vast amounts of data and identifying patterns that may not be apparent to human researchers, AI-powered systems can provide valuable insight into potential treatment options that may have been overlooked through traditional research methods. Our study underscores the significance of interdisciplinary collaboration between computational and experimental scientists in drug discovery and in developing a robust pipeline to identify potential leads for drug development.

Keywords: neurodegenerative disorders; Neuroinflammation; CXCR4; molecular docking; machine learning; applicability domain

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Introduction

Neurodegenerative disorder refers to an array of diverse conditions that result in progressive degeneration and loss of neurons, leading to a decline in cognitive and motor functions [1, 2]. There is a severe lack of effective diagnostic and treatment options for neurodegenerative disorders due to the complexity of the molecular mechanisms of their pathophysiology and the heterogeneity of the patient population [3]. Machine learning (ML), a branch of artificial intelligence (AI) and computer science, involves the use of data and algorithms to learn and make predictions [4]. This can be used for the development of approaches for the diagnosis and treatment of complex conditions, including neurodegenerative disorders, such as AD and PD [5]. Due to the availability of large amounts of experimental data, AI and ML have shown potential to be an indispensable tool to derive valuable insights and help in decision-making during drug development [6]. Thus, advancements in AI/ML algorithms have presented an unprecedented potential to accelerate drug development through AI/ML-driven approaches [7].

In our previous study, we observed the upregulation of the CXC motif chemokine receptor 4 (CXCR4), a member of the G protein-coupled receptor (GPCR) protein superfamily, in AD and PD, by analysing gene expression patterns of AD and PD patients compared to healthy controls of similar age, using publicly available RNA sequencing datasets [8]. Other studies have also reported upregulation of the CXCR4 gene in neurodegenerative disorder patients, including AD and PD, as compared to healthy age-matched controls [9, 10]. CXCR4 and its ligand CXC motif chemokine ligand 12 (CXCL12) are usually associated with bone marrow niche and haematopoiesis, the production of blood cellular components. However, CXCL12 is also expressed in various other tissues, like the brain, kidney, lung, heart, *etc.* Similarly, CXCR4 signalling pathways are crucial for various cellular processes, such as cellular proliferation and transport [11]. It has been suggested that dysregulation in the expression of CXCR4 and related microglial genes may play a role in age-related neurodegenerative disorders, including AD and PD [12].

In this study, we aim to find natural compounds that can potentially be used for the treatment of age-related neurodegenerative disorders, like AD and PD, through inhibition of the CXCR4 signalling pathway. Natural compounds have become increasingly more popular as lead compounds for drug development providing an alternative to synthetic compounds due to their biocompatibility and neuroprotective properties [13]. Natural compounds have been suggested to offer protection against excitotoxicity, neuroinflammation, oxidative stress and proteinopathies [14, 15]. Various compounds, such as curcumin, quercetin, resveratrol, ginsenosides and rosmarinic acid, are commonly found in plant species used in traditional medicine or food products like fruits, herbs and spices. However, their poor bioavailability and delivery in the central nervous system (CNS) limits their utilisation for therapeutic purposes. For example, even though curcumin has proven benefits in the treatment of AD, it is limited due to low absorption and bioavailability [16]. New pharmaceutical strategies like nanoparticles and nanocarriers are required for their targeted delivery and controlled release, to increase blood–brain barrier (BBB) permeability and improve stability against metabolic degradation, resulting in elevated bioavailability and reduced toxic side effects [17].

Materials and methods

Data curation

This study utilises the collection and analysis of the data from the Collection of Open Natural Products (COCONUT) Database [18] and the Binding Database (BindingDB) [19]. The COCONUT Database (http://www.coconut.naturalproducts.net), a large public resource of natural products, was the test data source. To enhance the relevance of the test data, the molecular structure of Mavorixafor, a BBB permeable inhibitor of CXCR4, was obtained from PubChem (https://www.pubchem.ncbi.nlm.nih.gov/compound/amd-070) and used as a reference to select 10 000 structurally similar compounds (Tanimoto coefficient > 65%) from the COCONUT Database. Similarity Search of COCONUT Database ranks compounds according to S_{AB} similarity score (defined as similarity between compound A and B) and produces a list of compounds with S_{AB} score greater than threshold (0.65 in our case). This search produced 10 000 compounds (maximum possible) with the last compound having a similarity score of 65.04%. In our study, we have used a relatively low threshold of 0.65 to increase chemical search space as done in previous studies [20]. A threshold of 0.65 was also evaluated to be most optimal for the prediction of compound-protein interaction in another study [21]. Simultaneously, the training data containing 1529 compounds with known activity against CXCR4 was sourced from the BindingDB (https://www.bindingdb.org). RDKit (https://www.rdkit.org), used for curation of this data, is an open-source cheminformatics and machine learning toolkit initially developed by Rational Discovery and presently maintained at Novartis Institutes for BioMedical Research [22]. Utilising the RDKit toolkit, Simplified Molecular Input Line Entry System (SMILES) strings of test and training data were standardised into canonical SMILES. This step was crucial in ensuring a consistent and comparable format for the chemical structures within the training and test data. Redundant and duplicate entries were removed from the training data. This resulted in final high-quality training data containing 1266 compounds (Table S1). These compounds were then classified according to their inhibitory activity as inhibitors ($IC50 \leq 100$ nM) and non-inhibitors ($IC50 > 100$ nM), for subsequent predictive modelling.

Blood–brain barrier permeability prediction and ADME analysis

Assessing a compound's potential to traverse the blood–brain barrier (BBB) is crucial for developing drug candidates targeting the CNS [23]. Our analysis of BBB permeability employed three tools—DeePred-BBB, LightBBB, and SwissADME. DeePred-BBB, utilising SMILES notations, predicted BBB permeability using a deep neural network (DNN)-based model, boasting superior accuracy (98.07%) compared to one-dimensional convolutional neural network (97.44%) and convolutional neural network (97.61%) [24]. LightBBB, a publicly available web server, also predicted BBB permeability through the SMILES query format [25]. SwissADME assessed physicochemical properties, pharmacokinetics, drug-likeness, and medicinal chemistry friendliness [26]. Compounds identified as BBB+ by all three tools underwent further evaluation of their absorption, distribution, metabolism, and excretion (ADME) properties. Drug-likeness, gauging the resemblance of a compound to existing drugs, is crucial for the drug development process [27]. Medicinal chemistry friendliness is essential to

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identify fragments that could impede drug development. Pan assay interference compounds (PAINS) denote substances with substructures leading to false positive biological outcomes [28]. While, Brenk alerts point to fragments with potential toxicity, reactivity, or properties detrimental to pharmacokinetics. Compounds with zero violations for drug-likeness filters (Lipinski, Ghose, Veber, Egan, and Muegge) and zero medicinal chemistry friendliness alerts (PAINS and Brenk) constituted the test dataset for machine learning analysis, focusing on their inhibitory activity against CXCR4.

Machine learning

Our approach was aimed at using machine learning methodologies to classify our test compounds as either inhibitors or non-inhibitors of CXCR4. For this classification process, we constructed training and test datasets.

Training dataset

The training dataset was constructed by retrieving compounds with known CXCR4 activity from BindingDB. To analyse the molecular characteristics of these compounds, we employed RDKit, a sophisticated and versatile cheminformatics toolkit. Molecular descriptors, quantitative representations of chemical features, were calculated for each compound within the training dataset, providing an understanding of their structural and physicochemical properties. Feature selection methods were employed utilising SciPy [29] and Scikit-learn [30] Python libraries to identify the most relevant features for machine learning. Initial screening involved discarding descriptors with more than 50% zeroes, enhancing the efficiency and relevance of subsequent analyses. Additionally, outliers were eliminated according to the z-score values of their features using a threshold of 3 standard deviations above and below the mean. Subsequently, we examined the relationships between different descriptors by calculating Pearson correlation coefficients using the pearsonr function from the scipy.stats module of SciPy. Pairs exhibiting correlations beyond a certain threshold underwent refinement, with one descriptor from each pair being eliminated. Visualisation of these correlations was achieved through the utilisation of Seaborn [31], a powerful Python library adept at rendering statistical data into insightful heatmaps. Finally, the Recursive Feature Elimination (RFE) function from *sklearn.feature_selection* module of Scikit-learn was utilised to rank the remaining descriptors based on their importance as features for predicting the activity of natural compounds. After the selection of the most relevant features, the Lazy Predict Python library (https://lazypredict.readthedocs.io/en/latest) was used to identify and select the most suitable models appropriate for our training data, optimising the subsequent phases of analysis. With our chosen models in place, we proceeded to train them using the training dataset.

Test dataset

The test dataset containing BBB+ natural compounds with desirable ADME properties was created from compounds retrieved from the COCONUT Database. DeePred-BBB, LightBBB, and SwissADME were used to select BBB+ compounds. Subsequently, SwissADME was used to evaluate violations for drug-likeness filters (Lipinski, Ghose, Veber, Egan, and Muegge) and zero medicinal chemistry friendliness alerts (PAINS and Brenk) and compounds with zero violations were selected for further processing. Just like the training dataset, RDKit was employed to analyse the molecular descriptors of these test compounds. SciPy Python libraries were again employed to verify that features obtained using the training dataset are also relevant to our training data. Relationships between different features in the test dataset were also evaluated by calculating Pearson correlation coefficients using the pearsonr function from the scipy.stats module of SciPy. Pairs exhibiting correlations beyond a certain threshold underwent refinement, with one descriptor from each pair being eliminated. Visualisation of these correlations was achieved through the utilisation of Seaborn. The ranges of each feature in the training dataset were then calculated and test molecules outside this range were eliminated. This process is essential to ensure the reliability of machine learning predictions. Thus, the remaining compounds constituted the test dataset for machine learning analysis. Finally, natural compounds within the test dataset were classified with selected training models as either inhibitors or non-inhibitors of CXCR4.

Performance metrics of selected models

For classification tasks, the results can be summarised into four categories, depending on their true and predicted labels [32]:

1. True Positives (TP) are compounds with the true and predicted labels as 1.
2. True Negatives (TN) are compounds with the true and predicted labels as 0.
3. False Positives (FP) are compounds with the true label as 0 and the predicted label as 1.
4. False Negatives (FN) are compounds with the true label as 1 and the predicted label as 0.

In our case '1' was the label given to the inhibitor and '0' was the label given to the non-inhibitor. The following performance metrics were calculated using the above four categories:

1. Accuracy: A fraction of the compounds correctly predicted.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

2. Precision: A fraction of the positively predicted compounds that are indeed positive.

$$Precision = \frac{TP}{TP + FP}$$

3. Recall: A fraction of positive compounds correctly predicted as positive.

$$Recall = \frac{TP}{TP + FN}$$

4. Specificity: A fraction of negative compounds correctly predicted as negative.

$$Specificity = \frac{TN}{TN + FP}$$

5. F_1 score: Harmonic mean of Precision and Recall.

$$F_1 = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}} = \frac{2TP}{2TP + FP + FN}$$

6. Area under the receiver operating characteristic curve (ROC AUC).

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Molecular docking analysis

To predict the predominant binding mode of identified natural compounds and estimate their binding affinity as inhibitors, we conducted their molecular docking analyses with CXCR4 utilising Vina-GPU 2.0 [33]. Using graphics processing units (GPUs), Vina-GPU 2.0 accelerates molecular docking techniques such as AutoDock Vina, QuickVina 2, and QuickVina-W, to facilitate a speedy analysis. Our specific choice, Vina-GPU+, optimises the acceleration of Vina-GPU and facilitates batch docking for multiple ligands against a single receptor. The X-ray crystal structure (resolution: 2.50 Å) of our target protein, CXCR4 (PDB Code: 3ODU), was obtained from Protein Data Bank (https://www.rcsb.org) [34]. To ensure precision in our analyses, the CXCR4 protein structure was optimised to remove water molecules, ligands, and cofactors. Hydrogen atoms were then added to the cleaned protein structure using Avogadro software [35]. In the preparation of ligands for docking, Open Babel was employed to convert the Structure Data Format (SDF) files to the MOL2 format [36]. Subsequently, the refined CXCR4 protein was subjected to molecular docking with Mavorixafor, a reference compound, and the test compounds to evaluate their binding affinities. The determination of potential inhibitors rested on the identification of compounds exhibiting high binding affinities. The interactions between natural compounds and specific residues of the CXCR4 protein were recorded and compared with existing literature. Identification of natural compounds with interactions in line with previous studies increases the reliability of the study.

Applicability domain

The applicability domain, in machine learning, denotes the part of the chemical space where the model can give reliable predictions for compounds present [37]. Various methods have been developed to evaluate the reliability of the machine learning predictions, and include range-based, distance-based, descriptor-based, and fragment-based approaches [38]. Mahalanobis distance (MD) is one such distance-based metric that accounts for correlations between different variables making it particularly effective in multivariate scenarios and is, therefore, widely used for applicability domain assessment [39]. It is defined as the measure of distance between a compound and the distribution of compounds in the training dataset. The calculation of MD is done by computing the mean vector of training data (consisting of the mean of all molecular descriptors) and the inverse of the covariance matrix (which represents the covariance between the descriptors). This allows us to know how far a compound is from the centre or mean of training compounds in a multidimensional chemical space created from molecular descriptors while considering the correlation between different descriptors, explaining how similar a compound is compared to training data [40]. In cheminformatics, it is significantly valuable for determining the similarity of compounds to the training data, which allows it to be used for assessment of the applicability domain. The aim is to find a similarity threshold that can be used to predict the inhibitory activity of new compounds reliably. For this purpose, the applicability domain was defined by the calculation of the average (d) and standard deviation (std) of MD of all compounds in the training dataset. These values were calculated utilising the *mahalanobis* function of *scipy.spatial.distance* module from the SciPy library. The threshold (t) to determine the border of the applicability domain was set as:

$$Threshold\ (t) = Distance\ (d) + Standard\ Deviation\ (std) \times z$$

Where z is an arbitrary parameter (default value is 0.5) [39]. Since MD accounts for the scale, correlation and shape of variables, it is considered better than Euclidean and Manhattan distance-based methods for assessment of the applicability domain. Using compounds structurally similar to Mavorixafor, a compound present in the training dataset, also improved their applicability to the training model. Furthermore, test data outside the descriptor range of training data was removed before activity prediction to remove compounds that may fall outside the applicability domain. The overall methodology of our study, encompassing data retrieval, preparation, machine learning, molecular docking and applicability domain analysis, has been elucidated in detail in Fig. 1.

Results

Blood–brain barrier permeability and ADME analysis

The significance of structural similarity is fundamental in the domain of drug design and discovery. This significance is based on the core assumption that compounds with similar structures are likely to manifest similar biological activities [41]. In the specific context of our study, our primary objective was to identify natural compounds sharing structural similarity with Mavorixafor, an inhibitor of CXCR4 that is permeable through the BBB. Thus, we retrieved 10 000 natural compounds (from the COCONUT Database) exhibiting a structural similarity exceeding 65% (using the Tanimoto coefficient) to Mavorixafor (Table S2). Since the goal of our study is to identify CXCR4 inhibitors for the treatment of neurodegenerative disorders, an important criterion of our selection process was the BBB permeability of compounds as this barrier selectively allows only certain molecules to penetrate the brain while preventing the transport of neurotoxins [42, 43]. To address this, we analysed all 10 000 natural compounds using three distinct tools (DeePred-BBB, LightBBB, and SwissADME) and selected 2712 compounds that were predicted to be BBB+ by all three tools (Table S3). Subsequently, these compounds were analysed for their pharmacokinetic properties. The parameters of ADME serve as pivotal metrics in comprehending a drug's pharmacokinetics [44]. ADME describes how our body processes a drug while it executes its therapeutic function and, thereby, provides an understanding of its molecular mechanisms, toxicity and drug–drug interactions [45, 46]. In an era before the formal integration of ADME testing into the drug discovery pipeline, a staggering 40% of compounds failed due to suboptimal pharmacokinetic properties. Thanks to advancements in this field, the failure rate has been now significantly reduced to less than 10% [47]. SwissADME was used to evaluate the ADME properties of BBB+ compounds, and those found to violate any of the five drug-likeness filters (Lipinski, Ghose, Veber, Egan, and Muegge) were systematically excluded. These are widely accepted guidelines in drug discovery and development to evaluate 'drugability', the ability of a compound to be used commercially as a pharmaceutical drug [48]. Similarly, compounds exhibiting one or more alerts for medicinal chemistry unfriendliness filters (PAINS and Brenk) were also eliminated. These filters are used to identify fragments or substructures of a compound that can lead to unfavourable chemical properties such as toxicity or non-specificity [49]. The removal of these compounds with poor pharmacokinetic properties is vital for increasing the reliability of the study. Thus, in the final refinement, 1566 compounds devoid of drug-likeness violations and medicinal chemistry

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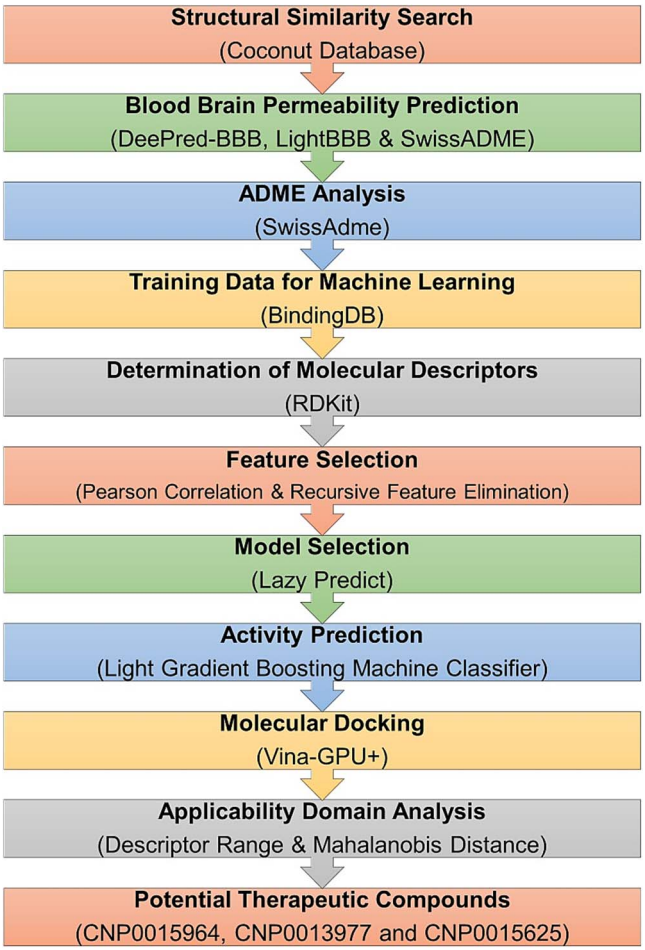


Figure 1. Methodology for prediction of CXCR4 inhibitors using machine learning. We initiated the process by extracting 10 000 natural compounds structurally similar to Mavorixafor from the COCONUT database, focusing on a Tanimoto coefficient greater than 0.65. Subsequent assessments included evaluating the BBB permeability using DeePred-BBB, LightBBB, and SwissAdme, followed by ADME analysis to evaluate drug-likeness and medicinal chemistry friendliness. Compounds with zero violations for different drug-likeness filters and zero medicinal chemistry alerts were selected to prepare the test dataset. On the other hand, compounds with known activity against CXCR4 were retrieved from BindingDB and used to prepare a training dataset. Molecular descriptors for both the test dataset and training dataset were calculated using RDKit. Less important descriptors were eliminated using feature selection methods. Lazy predict aided in selecting the best machine learning model, light gradient boosting machine (LGBM) classifier. Applying this model to the test dataset enabled the estimation of CXCR4 inhibitor activity. Compounds predicted to be inhibitors were evaluated for binding affinity through molecular docking and analysed for applicability domain using descriptor ranges and Mahalanobis distance. Compounds with the highest docking scores and presence inside the applicability domain were selected as potential therapeutic compounds.

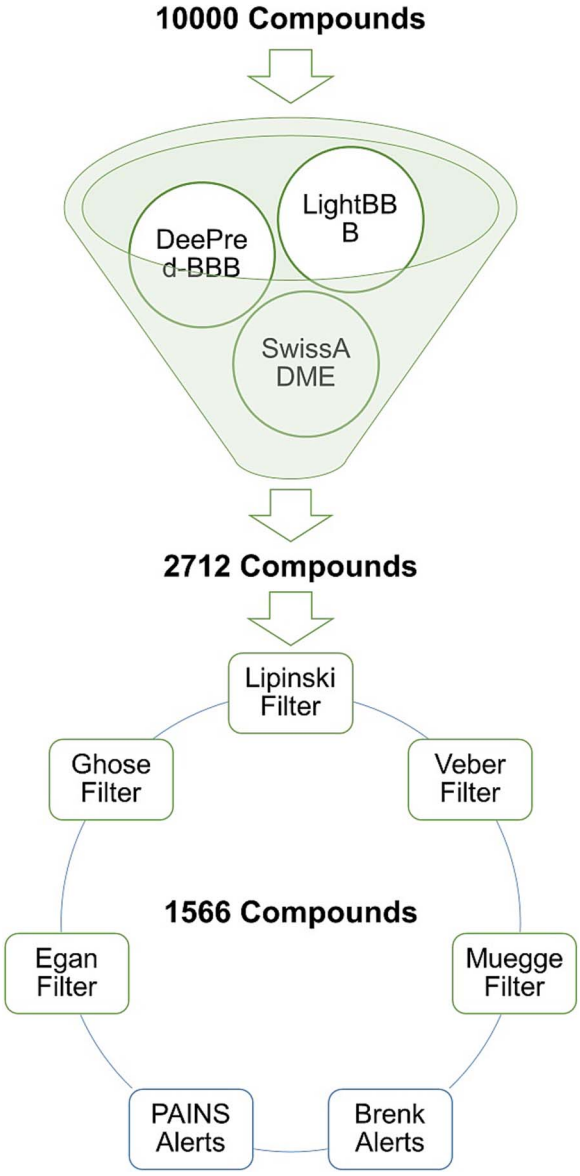


Figure 2. Prediction of blood–brain permeability and subsequent ADME analysis were crucial steps of our study. 10 000 compounds obtained from the COCONUT database were evaluated for their BBB permeability using three distinct tools, namely DeePred-BBB, LightBBB, and SwissADME, to ensure a comprehensive assessment. Subsequently, 2712 compounds emerged as promising candidates, being predicted as BBB+ by all three algorithms. The selected compounds underwent further scrutiny to determine their drug-likeness and medicinal chemistry friendliness. The compounds were evaluated using five drug-likeness filters (Lipinski, Ghose, Veber, Egan, and Muegge) and two medicinal chemistry filters (PAINS and Brenk). Following this analysis, 1566 compounds displayed zero violations for the drug-likeness filters and zero alerts for medicinal chemistry filters. Thus, compounds with positive blood–brain permeability, optimal drug-likeness and favourable medicinal chemistry attributes were selected for further analysis.

unfriendliness alerts, were selected to form the test dataset of natural compounds (Fig. 2 and Table S4).

Calculation of molecular descriptors and feature selection

To predict the activity of compounds against CXCR4, we created a training dataset using compounds retrieved with known activity against CXCR4 from the BindingDB, recording their IC50 values and SMILES notations. After data curation that involved standardisation of data and removal of duplicates 1266 compounds were left in training data. Compounds with IC50 values ≤100 nM were categorised as inhibitors, while those with IC50 values >100 nM were labelled as non-inhibitors. Recognising the intrinsic link between a compound's molecular structure and its biological activity [50], we evaluated molecular descriptors — depicting various physical and chemical characteristics — to enhance the efficiency of our drug design and discovery process. Utilising the Descriptors module of RDKit, an open-source tool for cheminformatics and machine learning, we calculated descriptors for both the training and test datasets (Tables S5 and S6). To increase the speed and accuracy of machine learning model creation,

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redundant data was removed, and most crucial features were selected for machine learning. Firstly, descriptors with more than 50% zero values were eliminated from either dataset, followed by the removal of outliers.

In the training dataset, any outlier with a z-score of 3 standard deviations above and below the mean for any feature was removed, finally resulting in 674 compounds in the training data, which included 322 inhibitors and 352 non-inhibitors (Table S7). Subsequently, we scrutinised the remaining descriptors for significant correlations, computing the Pearson correlation coefficient. If a pair of descriptors had the Pearson correlation coefficient exceeding 0.60 in the training dataset, one of the descriptors was removed. This process led to the selection of 17 descriptors out of the original 209 for further analysis. The Recursive Feature Elimination (RFE) algorithm of Scikit-learn ranked these 17 descriptors based on their feature importance by eliminating the least important features one by one, resulting in a list of descriptors ranked by their significance. A threshold of 0.60 was selected by using Grid Search on different thresholds and using RFE to select the threshold with the best model accuracy. The heatmaps of the Pearson correlation coefficients of the top 15 descriptors ranked by RFE for both the training and test dataset was generated using the Seaborn Python library (Fig. 3).

Model selection and activity prediction

After the creation of training and test datasets, the Lazy Predict Python library was used to identify the most optimum machine learning algorithm, as it can fit and evaluate all the machine learning models contained in the Scikit-learn Python library with a minimal amount of coding in a single step. It is composed of a Lazy Classifier and Lazy Regressor to perform the classification and regression tasks, respectively. We used Lazy Classifier to find a model for the classification of natural compounds as CXCR4 inhibitors or non-inhibitors by building and evaluating 27 different machine learning models using our training dataset. We used Lazy Classifier to select the best models for different numbers of descriptors removing the lowest RFE ranked descriptors one by one. After building models for different numbers of descriptors, we compared the highest accuracy obtained with each set of descriptors. The highest accuracy obtained for each set of descriptors has been shown in Fig. 4A. Additionally, the names and accuracy of the top three models for different numbers of descriptors have been mentioned in Table 1. The highest model accuracy was achieved at the set of top 15 ranked descriptors for Light Gradient Boosting Machine (LGBM) Classifier with 81.94% model accuracy, followed by eXtreme Gradient Boosting (XGB) Classifier with 79.86% model accuracy and Random Forest (RF) Classifier with 77.48% model accuracy. All the models and their accuracy with the set of top 15 ranked descriptors have been shown in Fig. 4B.

Finally, hyperparameter tuning was performed for all three models to find the optimum set of parameters. LGBM Classifier again showed the best model accuracy of 0.8222 (or 82.22%) with the parameters:

colsample_bytree : 0.6, *learning_rate* : 0.1, **max**_*depth* : 7,

min_*child_samples* : 20

n_estimators : 100, *num_leaves* : 20, *subsample* : 0.6

XGB Classifier showed an accuracy of 0.8074 (or 80.74%) with the following parameters:

colsample_bytree : 0.8, *learning_rate* : 0.2, **max**_*depth* : 3,

n_estimators : 100, *subsample* : 0.7

Random Forest Classifier showed an accuracy of 0.8 (or 80.00%) with the following parameters:

max_*depth* : *None*, **min**_*samples_leaf* : 1, **min**_*samples_split* : 10,

n_estimators : 300

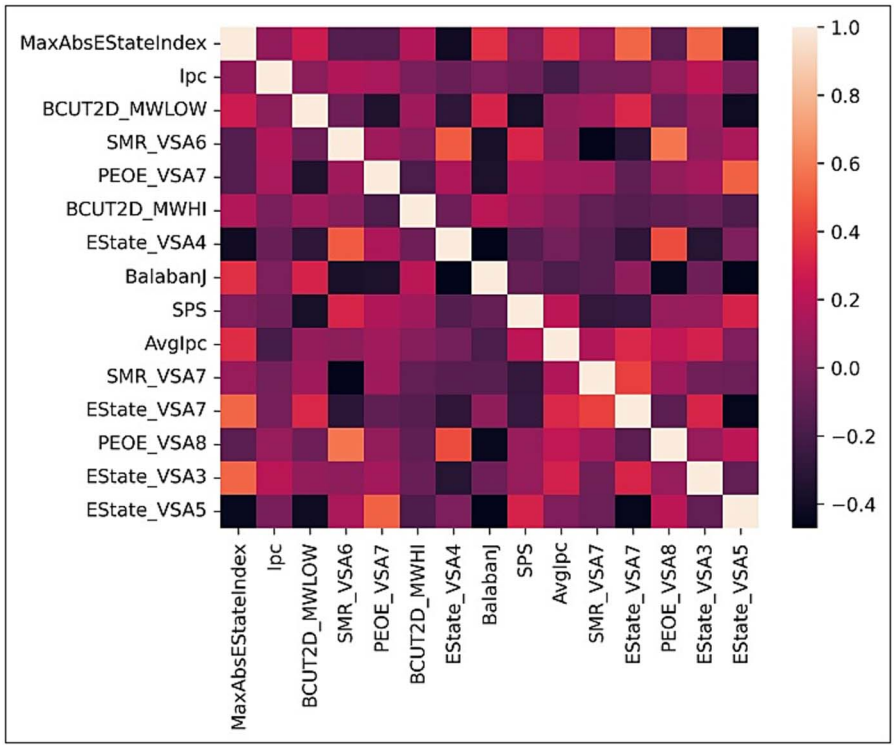
Along with model accuracy, these three models were also tested for various performance metrics such as Precision, Recall, Sensitivity, F1-score and ROC AUC Score. To calculate all these performance metrics, including accuracy, 20% of the training dataset was used as the validation dataset to perform holdout validation of data. In the holdout validation approach, training set is split into two parts, one for training (70%–80%) and one for validation (20%–30%). Subsequently, 70%–80% training set is used to tune the model and remaining 20%–30% is used to evaluate the model [51]. Precision signifies the fraction of positively predicted compounds that are actually positive. LGBM Classifier shows a Precision Score of 0.9 (or 90%). Specificity signifies the fraction of negative compounds that are correctly predicted as negative. The specificity of the LGBM Classifier was calculated to be 0.9048 (or 90.48%). Both of these metrics show that our model (LGBM Classifier) is highly accurate in classifying negatives as true negatives instead of false positives. Therefore, the LGBM Classifier was used for the prediction of the activity of test data using the top 15 ranked descriptors. Performance metrics (Accuracy, Precision, Recall, Sensitivity, F1-score and ROC AUC Score) of the top 3 models have been mentioned in Table 2.

Before using our trained model for activity prediction, it was important to remove outliers from our test data to only consider molecules within the descriptor space of training data to ensure consistency between both datasets. Thus, the range of features of training data was evaluated and outliers in the test data that were outside the range for corresponding features of training data were identified and removed. This was done to mitigate the data discrepancies between training and test datasets, thereby improving the reliability and applicability of the developed machine learning model. By aligning the range of features in both datasets and removing outliers that could adversely affect model accuracy, the model performance on unseen data is expected to improve. This resulted in 974 compounds in our test data that we can use for activity prediction (Table S8). Violin plots were generated using the Seaborn Python library to visualise feature distribution of test and training data (Fig. 5). The LGBM Classifier machine learning model was then used to classify our test compounds as inhibitors and non-inhibitors of CXCR4 (Table S9). Finally, 20 compounds that were classified as CXCR4 inhibitors were analysed for their binding affinity by molecular docking and applicability to the trained model through analysing the applicability domain.

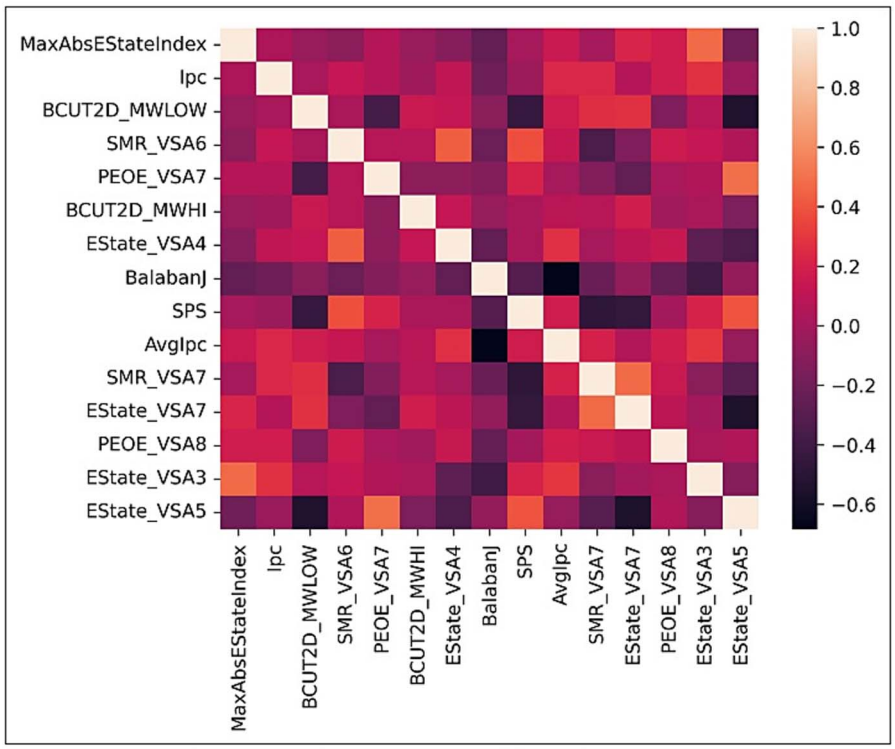
Validation: Molecular docking and applicability domain

After the LGBM Classifier was determined as the best model for our training data through Lazy Predict, test data was evaluated for their activity against CXCR4. 20 compounds, mentioned in Table 3, were predicted to possess inhibitory activity against CXCR4.

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A) Pearson Correlation Coefficient of Training Set



B) Pearson Correlation Coefficient of Test Set

Figure 3. Selection of molecular descriptors as best features for generation of machine learning model. Molecular descriptors were first evaluated for both the training dataset and test dataset. To eliminate less important descriptors, feature selection was performed. Descriptors with more than 50% zero values were eliminated. Pearson correlation test was performed for the rest of the descriptors and if the Pearson correlation coefficient of two descriptors was more than 0.60, one descriptor of the pair was eliminated. Finally, 17 descriptors were selected for further analysis. A heatmap of Pearson correlation coefficients was prepared for both a) the training dataset and B) the test dataset, using the top 15 descriptors, predicted by recursive feature elimination, that displayed maximum model accuracy.

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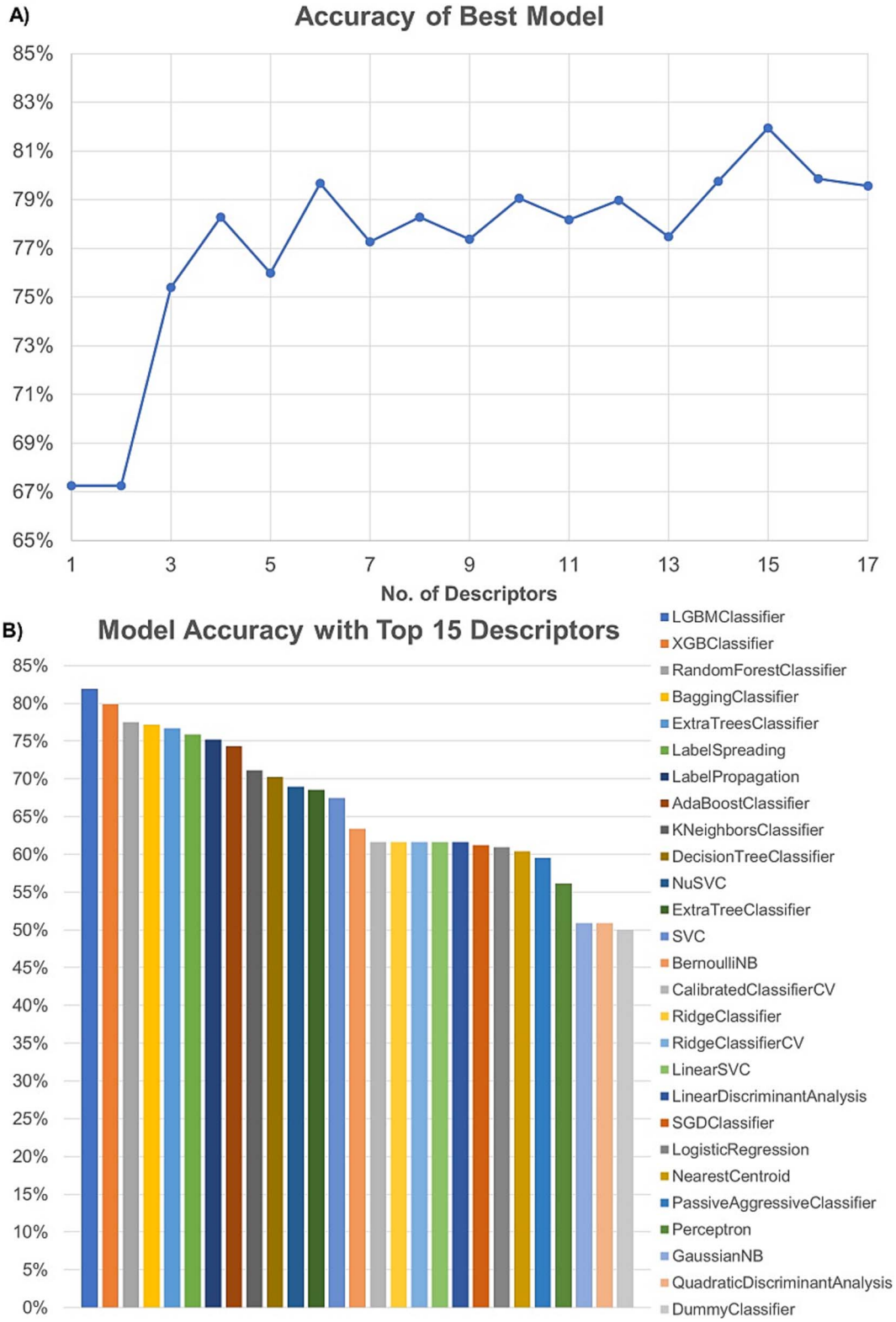


Figure 4. Selection of best machine learning model for classification of activity of natural compounds as inhibitors and non-inhibitors of CXCR4. 17 descriptors obtained after the Pearson correlation test were ranked by their feature importance using the recursive feature elimination (RFE) method to improve model performance by removing less important features. Lazy predict was used to select the best model for different numbers of descriptors removing lowest lowest-ranked descriptors one by one. Best model accuracy was achieved at 15 descriptors with 81.94% accuracy for the light gradient boosting machine (LGBM) classifier. A) Comparison of balanced accuracy of best models at the different number of descriptors. B) Comparison of accuracy of different models with a training dataset of 15 highest ranked descriptors.

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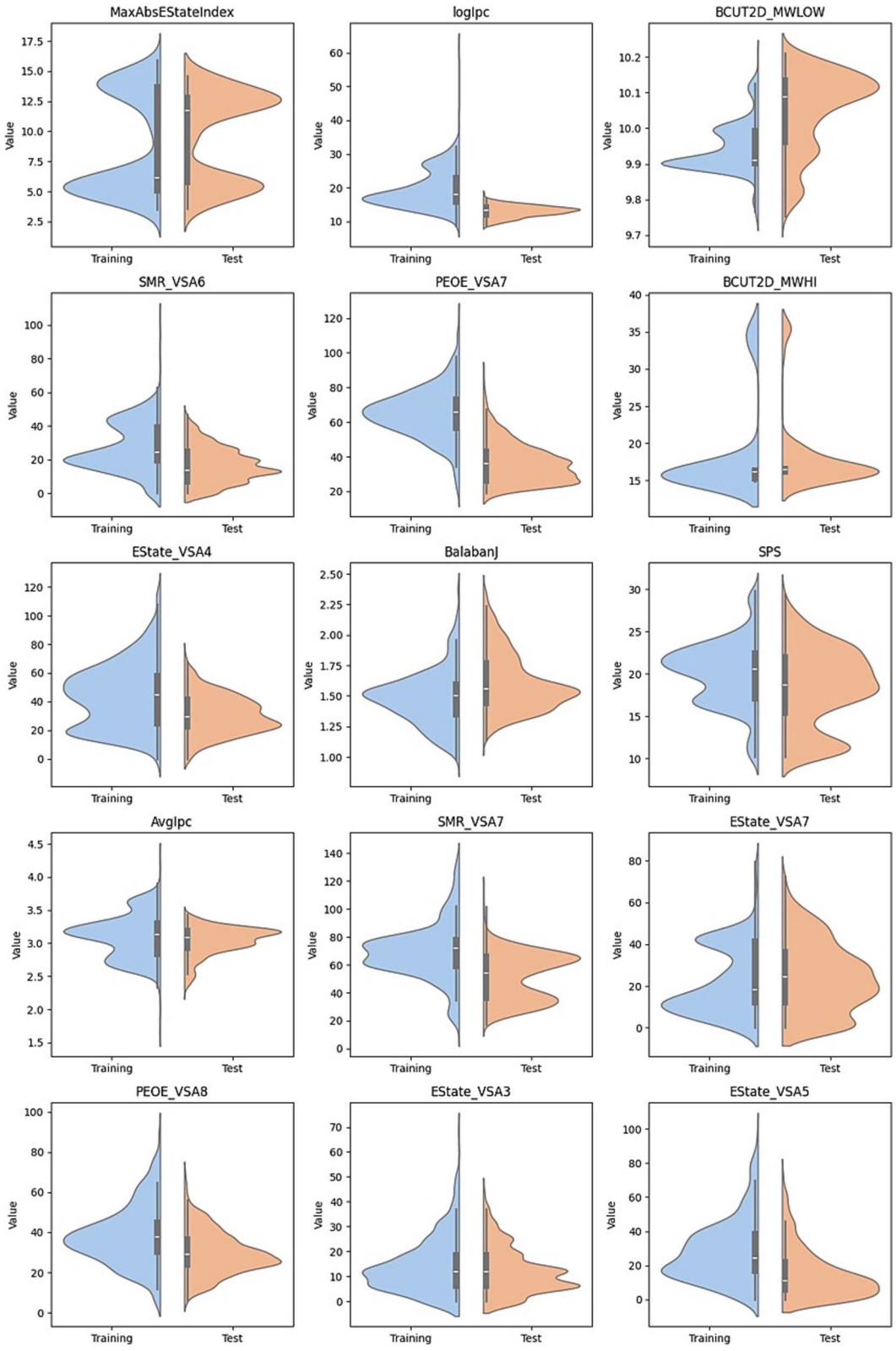


Figure 5. Feature distribution of training and test data. Outliers in test data that were outside the feature range of training data were removed. Subsequently, violin plots were generated for each feature to compare the distribution of the top 15 features (ranked according to RFE) in training and test data. Each subplot contains the individual feature, with the value of the feature mentioned in the y-axis and the data type (training or test) mentioned in the x-axis. The Ipc feature was transformed using a logarithmic scale to improve visibility. The feature distribution shows that the range of features in test data is within that of training data, ensuring the integrity of the predicted models.

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Table 1. Comparison of accuracy of top 3 models with different numbers of descriptors.

Features	Best Model	Accuracy	2nd Best Model	Accuracy	3rd Best Model	Accuracy
1	ExtraTreeClassifier	67.26%	RandomForestClassifier	67.06%	ExtraTreesClassifier	66.37%
2	ExtraTreeClassifier	67.26%	RandomForestClassifier	66.27%	KNeighborsClassifier	66.17%
3	ExtraTreesClassifier	75.40%	XGBClassifier	75.10%	DecisionTreeClassifier	73.81%
4	ExtraTreesClassifier	78.27%	LGBMClassifier	76.79%	RandomForestClassifier	76.69%
5	LGBMClassifier	75.99%	RandomForestClassifier	75.89%	ExtraTreesClassifier	75.00%
6	ExtraTreesClassifier	79.66%	RandomForestClassifier	76.79%	XGBClassifier	76.79%
7	XGBClassifier	77.28%	ExtraTreesClassifier	75.89%	LGBMClassifier	75.10%
8	LGBMClassifier	78.27%	XGBClassifier	78.17%	ExtraTreesClassifier	76.59%
9	LGBMClassifier	77.38%	RandomForestClassifier	75.30%	ExtraTreesClassifier	73.61%
10	RandomForestClassifier	79.07%	LGBMClassifier	76.79%	XGBClassifier	74.40%
11	RandomForestClassifier	78.17%	LGBMClassifier	75.99%	XGBClassifier	75.30%
12	RandomForestClassifier	78.97%	LGBMClassifier	77.48%	ExtraTreesClassifier	77.18%
13	LGBMClassifier	77.48%	RandomForestClassifier	77.38%	XGBClassifier	76.79%
14	LGBMClassifier	79.76%	XGBClassifier	79.17%	RandomForestClassifier	76.49%
15	**LGBMClassifier**	**81.94%**	XGBClassifier	79.86%	RandomForestClassifier	77.48%
16	RandomForestClassifier	79.86%	LGBMClassifier	79.76%	ExtraTreesClassifier	76.59%
17	LGBMClassifier	79.56%	XGBClassifier	78.47%	RandomForestClassifier	78.08%

Table 2. Performance metrics of top 3 models selected through lazy predict and hyperparameter tuning.

Classifier	Parameter	Accuracy	Precision	Recall	Specificity	F1 Score	ROC AUC Score
Light Gradient-Boosting Machine (LGBM)	colsample_bytree: 0.6 learning_rate: 0.1 max_depth: 7 min_child_samples: 20 n_estimators: 100 num_leaves: 20 subsample: 0.6	0.8222	0.9000	0.7500	0.9048	0.8182	0.8274
Extreme Gradient Boosting (XGB)	colsample_bytree: 0.8 learning_rate: 0.2 max_depth: 3 n_estimators: 100 subsample: 0.7	0.8074	0.8594	0.7639	0.8571	0.8088	0.8105
Random Forest (RF)	max_depth: None min_samples_leaf: 1 min_samples_split: 10 n_estimators: 300	0.8000	0.8571	0.7500	0.8571	0.8000	0.8036

Mavorixafor and 20 compounds classified as CXCR4 inhibitors, by LGBM Classifier, after machine learning were docked to evaluate their binding affinity with CXCR4 using Vina-GPU+. It is one of the three docking methods of Vina-GPU 2.0 and allows accelerated docking using a GPU and batch processing of multiple ligands with a single receptor molecule. Our reference drug (Mavorixafor) had a binding score of −7.4 kcal/mol. 19 out of 20 compounds exhibited better docking scores than Mavorixafor, ranging from −7.5 kcal/mol to −11.1 kcal/mol. One compound had a slightly lower binding score (−7.3 kcal/mol) than Mavorixafor. The next step was to analyse these compounds for their applicability to the trained model. Mahalanobis distance (MD) was used as a metric for this purpose, as it allows us to check whether a compound falls in the descriptor space of training data. The applicability domain was defined by the calculation of the average (d) and standard deviation (std) of MD of all compounds in the training data. These values were calculated utilising These values were calculated utilising *mahalanobis* function of *scipy.spatial.distance* module from the SciPy library. The threshold (t) to determine the border of the applicability domain was set as:

$$threshold\ (t) = d + std \times z$$

For our training data, values of d, std and t were determined to be 3.63, 1.34 and 4.30, respectively. 7 compounds with MD less than 4.30 were, therefore, observed to fall in the applicability domain of our training data. However, one compound (CNP0399717) had a slightly lower binding score as mentioned above. All 20 compounds were visualised for their MD and their presence inside or outside the applicability domain using the pyplot interface of matplotlib [52], as shown in Fig. 6A. Compounds with green colour refer to those that are inside the applicability domain. While compounds with red colour refer to those that are outside the applicability domain.

Since we are dealing with 15 descriptors for our study, we used Principal component analysis (PCA) to reduce the dimensionality of our data while minimising information loss. This is done by the

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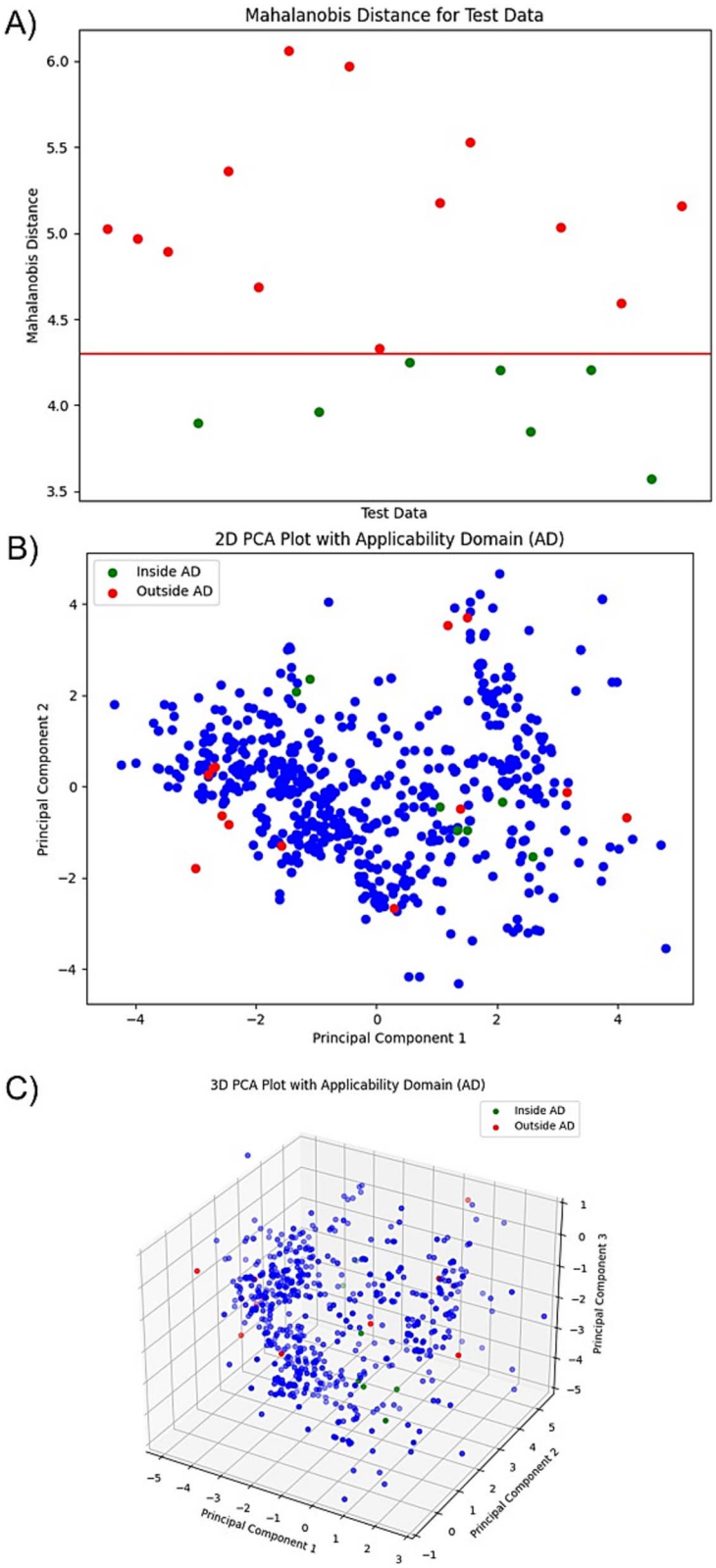


Figure 6. Applicability domain and PCA analysis. After lazy predict determined the top machine learning algorithms (LGBM classifier), compounds in the test dataset were evaluated for their activity against CXCR4. After the prediction of 20 compounds as inhibitors, these compounds were tested if they fell in the applicability domain. Mahalanobis distance was used as a metric to analyse the presence of our selected test compounds falling in the descriptor space of training data. To define the applicability domain, we calculated the average (d) and the standard deviation (std) of the Mahalanobis distance. The threshold was set as d + std x z, where z is the arbitrary parameter (default value is 0.5). Using this approach 7 compounds were observed to fall in the applicability domain. Compounds with values below this threshold refer to those that are inside the applicability domain. While compounds with values above this threshold refer to those that are outside the applicability domain.

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Table 3. List of 20 natural compounds predicted as CXCR4 inhibitors by LGBM classifier with their docking scores and their presence in the applicability domain.

S.No.	ID	Mahalanobis Distance	Within AD	Docking Score (kcal/mol)
1	CNP0004424	4.68	No	−11.1
2	**CNP0015964**	**3.89**	**Yes**	**−10.7**
3	CNP0319279	5.97	No	−10
4	CNP0094414	5.02	No	−9.8
5	CNP0165909	4.33	No	−9.7
6	**CNP0013977**	**3.57**	**Yes**	**−9.2**
7	CNP0450948	5.16	No	−9.1
8	**CNP0015625**	**4.20**	**Yes**	**−9**
9	CNP0014385	4.89	No	−8.8
10	**CNP0127043**	**4.25**	**Yes**	**−8.7**
11	CNP0015957	5.17	No	−8.5
12	**CNP0314177**	**4.20**	**Yes**	**−8.4**
13	CNP0015859	5.36	No	−8.3
14	CNP0408379	5.03	No	−8
15	CNP0360705	6.06	No	−7.8
16	CNP0397928	5.53	No	−7.8
17	CNP0122007	4.59	No	−7.7
18	CNP0216455	4.97	No	−7.6
19	**CNP0015823**	**3.85**	**Yes**	**−7.5**
20	**CNP0399717**	**3.96**	**Yes**	**−7.3**

creation of new uncorrelated variables, known as the principal components, defined according to the given data, which makes it an adaptive data analysis technique [53]. Depending on the number of principal components being selected while visualisation, we can generate 2D PCA with two principal components or 3D PCA with three principal components. 2D PCA and 3D PCA plots were visualised while using the pyplot interface of matplotlib [52] and predicted test molecules were observed in the PCA space of training data along with their status for the applicability domain in Fig. 6A and B, respectively. After accessing the compounds for applicability domain, their interaction with different residues of CXCR4 was analysed. CXCR4 protein contains seven transmembrane (TM) helices, TM1 to TM7 that are arranged in a barrel-like structure with the inside of the barrel being more hydrophilic and the outside being more hydrophobic [54]. Besides TM helices, CXCR4 contains an extracellular N-terminal domain, three extracellular loops (ECL), three intracellular loops (ICL) and an intracellular C-terminal domain [55].

Important interacting residues of CXCR4 are Glu32 in N-terminal domain; Phe36/Asn37/Leu41/Tyr45 in TM1; Trp94/Asp97/Ala98 in TM2; Val112/His113/Tyr116 in TM3; His281/Ile284/Ser285/Glu288 in TM7; Trp102 in ECL1; and Cys186/Arg188 in ECL2 [56]. Most of these residues can be seen as interacting residues of our selected compounds (Fig. 7). TRP94, VAL112 and HIS113 were observed to be the most prominent interaction residues across all the molecules. Thus, our results align with existing literature about the interaction of small molecules with CXCR4. CNP0015964 (benzimidazole derivative), CNP0013977 (pyridinylpiperazine derivative), and CNP0015625 (aminopyridine derivative) were the top 3 compounds with the highest docking scores that were within the applicability domain of our trained model. All the most prominent interacting residues, TRP94, VAL112 and HIS113, are observed to be interacting with CXCR4 in all three molecules (Fig. 7). Furthermore, they had 44.59%, 24.32% and 21.62% higher docking scores, respectively, as compared to Mavorixafor. Thus, these compounds can potentially be more effective inhibitors than Mavorixafor and may be used to treat neurodegenerative disorders.

Discussion

Several studies have reported that CXCR4, a chemokine receptor predominantly expressed on B-cell surface, has diverse regulatory functions within the immune system, cell proliferation, and neurodevelopment [55]. CXCR4 downregulation is a critical event that facilitates the migration of immature B cells from the bone marrow into the bloodstream (Fig. 8) [57]. However, in neurodegenerative disorders, an aberrant upregulation of CXCR4 has been implicated in a cascade of events leading to reduced B-cell migration, decreased antibody secretion, and the consequential aggregation of toxic proteins, such as Aβ in AD and α-Syn in PD (Fig. 8) [58, 59]. In the absence of effective treatment options, natural compounds have recently gained significant attention due to their inherent biocompatibility, low toxicity and neuroprotective properties, making them attractive candidates for therapeutic alternatives to synthetic chemical compounds [60, 61]. Notably, among their diverse protective effects, natural compounds can counter excitotoxicity, reduce neuroinflammation, alleviate oxidative stress, and mitigate proteinopathies [62].

The main objective of this study was to classify natural compounds as inhibitors (IC50 $\leq$ 100 nM) or non-inhibitors (IC50 > 100 nM) termed according to their activity (IC50 values) against CXCR4. This activity was predicted through cheminformatics-guided machine learning models based on the molecular descriptors data. This would ultimately allow their use as a potential therapeutic option for neurodegenerative disorders, particularly AD and PD. To create a test dataset, we selected 10 000 natural compounds from the COCONUT Database, ensuring that they were structurally similar (Tanimoto Coefficient > 65%) to Mavorixafor. We then screened these compounds for their ability to cross the BBB, an important factor for drug efficacy in CNS disorders [63]. Only 2712 compounds

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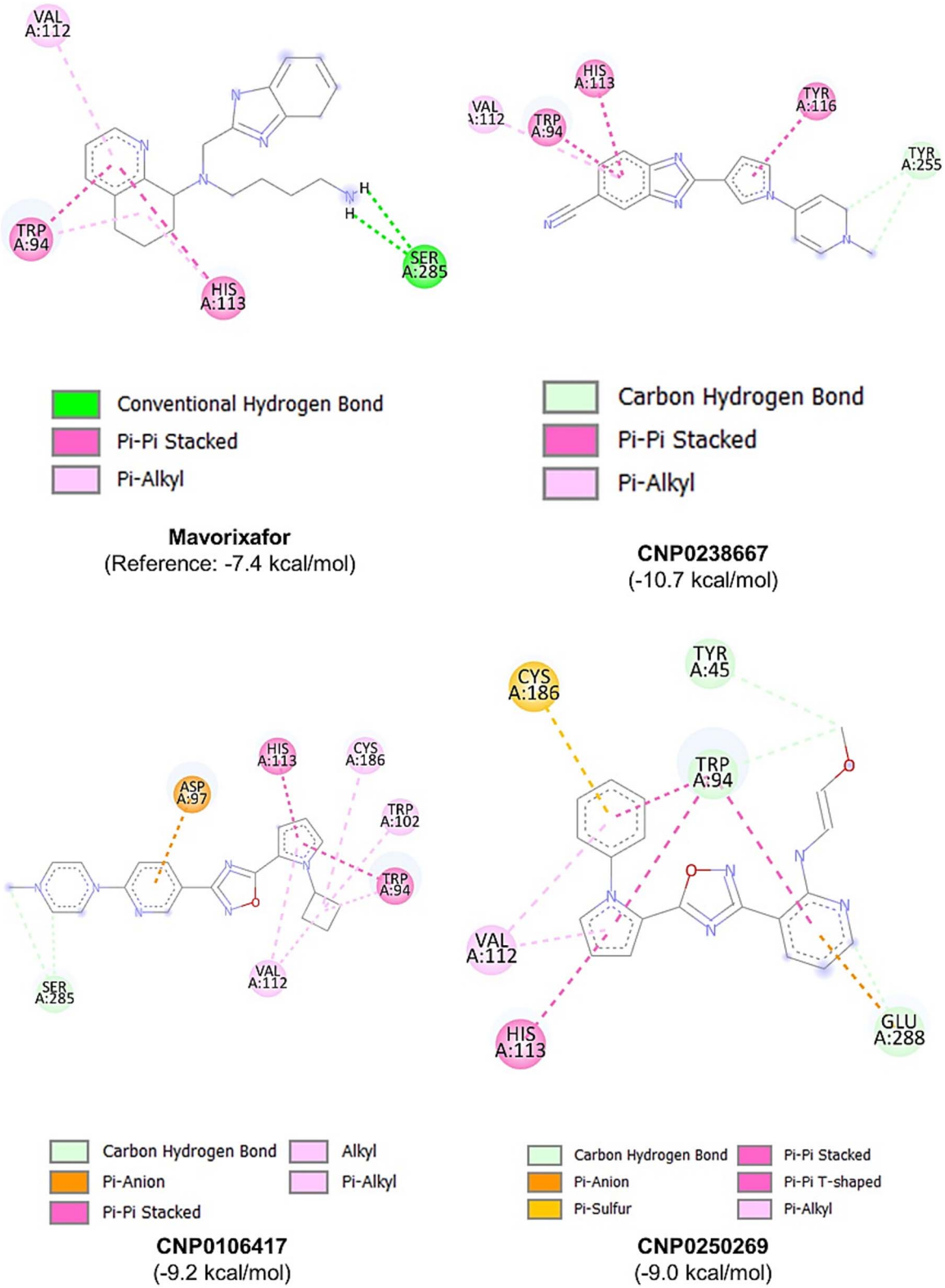


Figure 7. Receptor-ligand interactions of Mavorixafor and the top 3 compounds predicted by machine learning. Different types of binding interactions of the top 3 compounds and Mavorixafor with CXCR4 have been shown. Important interacting residues of CXCR4 are Glu32 in N-terminal domain; Phe36/Asn37/Leu41/Tyr45 in TM1; Trp94/Asp97/Ala98 in TM2; Val112/His113/Tyr116 in TM3; His281/Ile284/Ser285/Glu288 in TM7; Trp102 in ECL1; and Cys186/Arg188 in ECL2. Certain residues, like TRP94, VAL112 and HIS113, have been observed to be binding residues in all three selected molecules and Mavorixafor. These results about interacting residues complement existing literature.

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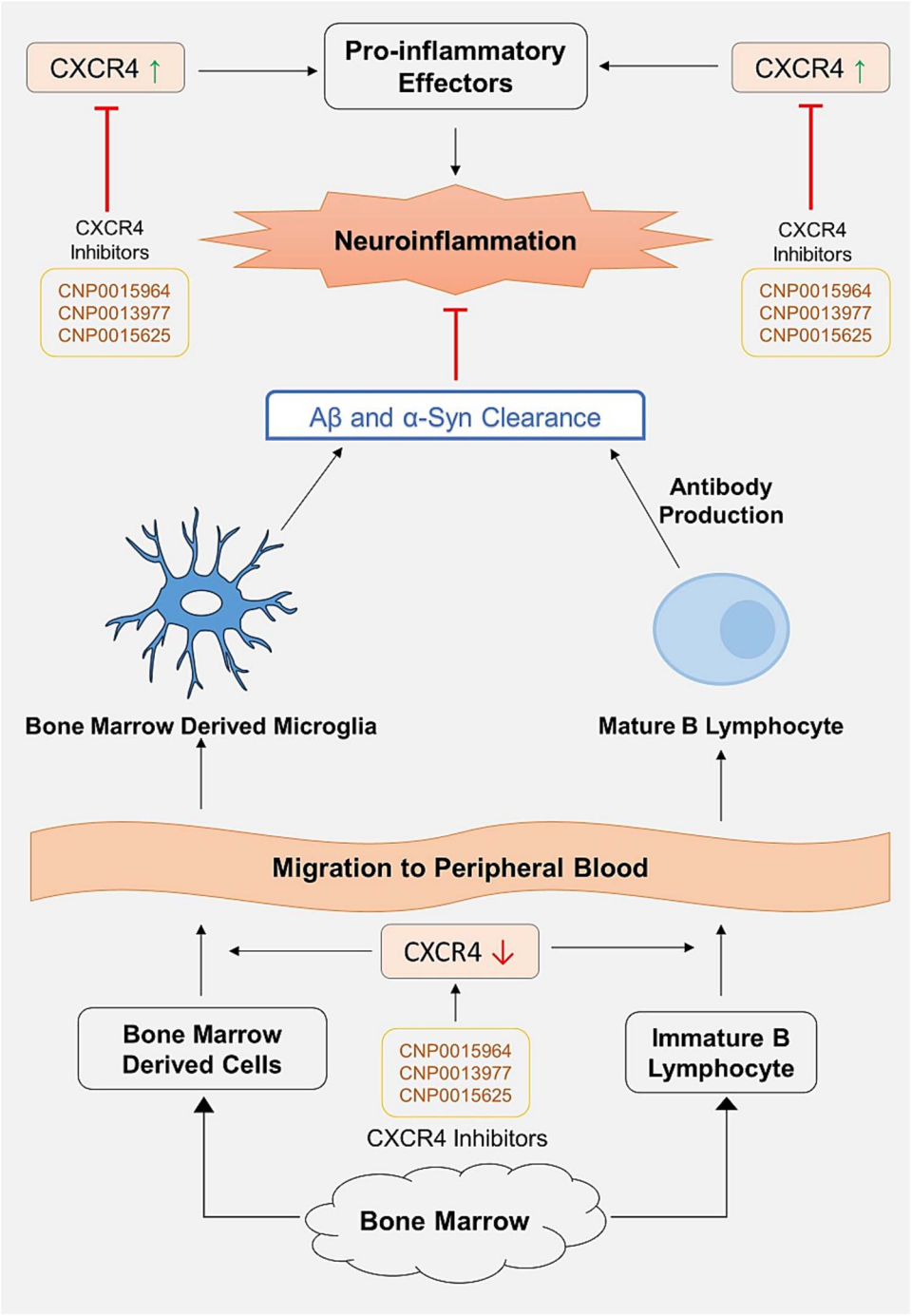


Figure 8. Fall in CXCR4 expression is vital for the migration of immature B lymphocytes and bone marrow-derived cells from the bone marrow into the peripheral blood. Mature B lymphocytes secrete antibodies that cross the blood–brain barrier and target toxic proteins, like Aβ and α-Syn. Inhibition of CXCR4 expression through natural compounds may be helpful for Aβ and α-Syn clearance, thereby helping in treating neurodegenerative disorders caused by aggregation of such toxic proteins, namely AD and PD. similarly, an increase in CXCR4 expression activates effectors (such as Akt, ERK, c-Jun, NF-kB, p38, mTOR, *etc.*), as it is a pro-inflammatory chemokine, and results in neuroinflammation, which is an important characteristic of the pathophysiology of neurodegenerative disorders, such as AD and PD. thus, CXCR4 inhibitors may help in regulating CXCR4 expression to normal range, thereby providing protection against neuroinflammation and cell death.

that passed the initial screening by DeePred-BBB, LightBBB, and SwissADME were selected for further analysis. We also assessed the drug-likeness and medicinal chemistry friendliness of these compounds to eliminate those with suboptimal ADME properties, which can hinder drug development [64]. This resulted in the identification of 1566 compounds that exhibited zero violations for Lipinski, Ghose, Veber, Egan, and Muegge filters, as well as zero Pan Assay Interference Compounds (PAINS) and Brenk alerts. These compounds were considered suitable for further investigation. Finally, we calculated the molecular descriptors of these compounds using RDKit software to create a test dataset for machine learning.

Simultaneously, a robust training dataset of 674 molecules was formulated using molecules with documented activity

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against CXCR4 from the BindingDB. Lazy Predict facilitated the exploration of various classification algorithms, ultimately leading to the selection of the LGBM Classifier, XGB Classifier and Random Forest Classifier, all of which demonstrated superior accuracy against the training dataset. LGBM Classifier offering the best model accuracy after hyperparameter tuning was ultimately used for predicting the activity of test compounds and identified 20 compounds as potential inhibitors (Table 3). Subsequently, we delved into a detailed assessment of the binding affinity of these 20 compounds employing Vina-GPU 2.0 and docked these compounds with CXCR4. Remarkably, 19 out of 20 compounds exhibited superior binding scores compared to Mavorixafor. The last compound had only a slightly inferior binding score compared to Mavorixafor. Finally, these compounds were tested for their applicability to our training model. For this purpose, Mahalanobis distance, a distance-based metric employed to define the applicability domain [39], was used to analyse the presence of predicted test compounds in the feature space of our training data. Furthermore, test compounds outside the feature range of training data were already eliminated before activity prediction, producing a test dataset of 974 compounds, reducing the likelihood of predicted test compounds falling outside the applicability domain of training data. Therefore, 7 compounds were observed to fulfil the applicability domain criteria and prediction for these compounds by our machine learning model can be suggested to be reliable.

Three compounds, CNP0015964 (benzimidazole derivative), CNP0013977 (pyridinylpiperazine derivative), and CNP0015625 (aminopyridine derivative), which emerged as the top three CXCR4 inhibitors were within the applicability domain of the model used for training and displayed higher docking scores than Mavorixafor by 44.59%, 24.32% and 21.62%, respectively. Many benzimidazole, pyridinylpiperazine and aminopyridine derivatives have been reported for their neuroprotective properties [65–67], suggesting the potential of these compounds as therapeutic options for neurodegenerative disorders. Analysis of the binding interactions between ligands and receptors revealed specific residues TRP94, VAL112, and HIS113 as potential determinants for CXCR4 inhibition, which have been reported in previous studies [8, 68, 69]. Additionally, it is worth mentioning that these residues were consistently observed as binding residues in all three selected molecules as well as Mavorixafor. These observations strengthen the idea that CNP0015964, CNP0013977, and CNP0015625 have therapeutic value as CXCR4 inhibitors and could be potentially effective in the management of neurodegenerative disorders (Fig. 8). The significance of our findings extends beyond the identification of potential therapeutic molecules as it not only unravels the structural motifs that influence CXCR4 inhibition but also establishes a robust computational framework that seamlessly integrates cheminformatics and machine learning to expedite the discovery of CXCR4 inhibitors from natural compound libraries. The cost-effectiveness and efficiency associated with this strategy position it as a compelling avenue for drug development, particularly in the context of inflammatory and immune-related disorders.

Conclusion

AI and ML algorithms have made significant advances over the past decade. These computational tools, along with the widespread availability of biological data, have become indispensable for drug design and discovery. They not only expedite the drug development process but also offer a cost-effective way of identifying more potent treatment options for various diseases, including neurodegenerative disorders. In this study, we utilised Lazy Predict Python libraries to semi-automate machine learning to classify natural compounds as either inhibitors or non-inhibitors of CXCR4. Our methodology integrated machine learning with structural similarity, ADME, and molecular docking, leading to the identification of optimal candidates for lead discovery. This versatile approach can be applied to other drug targets beyond CXCR4. By utilising existing data to generate machine learning models, our methodology accelerates the discovery of potential lead compounds for various targets. This not only improves the efficiency of the drug discovery pipeline but also offers a robust framework that can be used in conjunction with traditional experimental approaches. As the landscape of drug development continues to evolve, the amalgamation of computational and experimental techniques stands poised to redefine the efficiency and success of therapeutic interventions.

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Author contributions

P.K. and R.T. conceived and designed the manuscript. R.T. has collected, analysed, and critically evaluated these data as part of his PhD thesis. R.T. has prepared figures and tables. P.K. and R.T. analysed the entire data and wrote the manuscript.

Supplementary data

Supplementary data is available at *INTBIO Journal* online.

Conflict of interest: All authors have read the manuscript and declared no conflict or competing interests.

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